Context-Prioritized Hybrid AI for Real-Time Cyber and IoT Threat Detection: Integrating Supervised Trees, Subspace Autoencoders, Active Learning, and SHAP Explainability

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Abstract

Modern Security Operations Centers (SOCs) confront a persistent operational paradox: deterministic signature engines and static statistical rules fail to generalize to unseen zero-day attacks while generating overwhelming false-positive alert volumes during routine business-hour traffic surges. Conversely, standalone supervised machine learning classifiers exhibit poor recall on novel attack signatures absent from training corpora, whereas standalone unsupervised anomaly detectors suffer from high false-alarm rates when exposed to benign administrative bursts. This paper presents a four-stage, context-prioritized hybrid artificial intelligence architecture designed for real-time threat detection and risk prioritization across structured enterprise security and Internet of Things (IoT) telemetry. The proposed system extracts seven orthogonal temporal, spatial, and categorical flow features over sliding 5-second windows, enforces leakage-free chronological time-series validation, and executes a supervised gradient-boosted decision tree classifier in parallel with a 3-dimensional subspace reconstruction autoencoder. Candidate detections are subsequently filtered through dynamic diurnal thresholding, asynchronous threat-intelligence enrichment, human-in-the-loop active learning boundary suppression, and Configuration Management Database (CMDB) asset criticality weighting. Evaluated across a 12,000-flow multi-asset benchmark with 3,600 chronologically held-out test flows (including 140 unseen zero-day covert DNS/TLS exfiltration attacks), the complete architecture achieves **99.72% overall classification accuracy**, **97.1% precision**, **100.0% recall** (**98.5% F1-score**), and **100.0% zero-day recall** at an inference latency of **0.0020 ms per flow**. Operationally, the framework reduces false positives by **96.7%** (from 307 to 10), compresses the immediate Priority-1 (P1) analyst triage queue by **78.4%** (from 514 raw baseline alerts to 111 critical-asset incidents), and lowers Mean Time to Triage (MTTT) by **77.3%** (from 18.5 minutes to 4.2 minutes) via local SHapley Additive exPlanations (SHAP).

Keywords: Intrusion Detection System (IDS), Internet of Things (IoT) Security, Hybrid Ensemble Learning, Subspace Autoencoder, Contextual Risk Scoring, Active Learning, Explainable AI (SHAP).

1 Introduction and Operational Problem Formulation

Enterprise networks and Industrial Internet of Things (IIoT) deployments generate millions of structured flow transcripts, intrusion alerts, and cloud control-plane audit logs daily [4, 5, 6]. Security Operations Centers (SOCs) tasked with monitoring these heterogeneous environments face two compounding challenges. First, traditional deterministic detection mechanisms—such as static volumetric Z-score cutoffs, Sigma SIEM signatures [7], and YARA pattern rules [8]—are inherently retrospective. While effective against known indicators of compromise (IOCs), deterministic rules fail to detect low-and-slow zero-day covert channels, such as fixed-jitter Command-and-Control (C2) beaconing or encrypted DNS/TLS exfiltration. Second, static statistical thresholds trigger hundreds of false-positive alarms during legitimate diurnal business-hour traffic peaks (08:00–18:00) and scheduled enterprise server backups, resulting in severe analyst alert fatigue.

Furthermore, conventional anomaly detection research frequently evaluates machine learning models using randomized k-fold cross-validation and uniform alert weighting. Randomized splitting introduces temporal data leakage by scattering contiguous multi-packet attack sessions across both training and test partitions, artificially inflating reported accuracy. Uniform alert weighting treats an anomaly on a peripheral IoT smart bulb as operationally equivalent to an anomaly on a core Active Directory Domain Controller, misallocating scarce analyst triage capacity.

Core Research Contribution: This paper formulates and validates an end-to-end, four-stage hybrid AI pipeline that resolves the precision–recall trade-off between known and zero-day cyber threats. By coupling parallel supervised/unsupervised inference with diurnal threshold adaptation, threat-intelligence enrichment, active learning feedback, asset criticality scoring, and local SHAP attributions, the system elevates F1-score from 48.9% (deterministic baseline) to 98.5% and overall accuracy from 88.00% to 99.72%.

2 Related Work and Benchmark Telemetry Foundations

Empirical intrusion detection research relies on standardized network corpora that capture both normal background traffic and contemporary adversarial tactics. Moustafa and Slay introduced the UNSW-NB15 dataset [1], synthesizing nine modern attack categories alongside realistic normal network flows. Sharafaldin et al. developed the CICIDS2017 benchmark [2] at the Canadian Institute for Cybersecurity, providing labeled bidirectional flow features across brute-force, DoS/DDoS, web infiltration, and botnet vectors. To address resource-constrained smart devices, Koroniotis et al. constructed the BoT-IoT corpus [3], capturing over 72 million records of IoT reconnaissance, DDoS, keylogging, and data exfiltration.

In algorithmic threat modeling, supervised gradient-boosted decision trees—specifically XGBoost [9] and LightGBM [10]—demonstrate state-of-the-art precision on tabular flow representations of known attack families at sub-millisecond CPU inference speeds. For unsupervised zero-day anomaly detection, Liu et al. formulated the Isolation Forest algorithm [11], which isolates outliers via random recursive partitioning in linear time complexity, while deep and subspace autoencoders identify novel attacks via manifold reconstruction error. Finally, Lundberg and Lee introduced SHAP (SHapley Additive exPlanations) [13], providing a game-theoretic foundation for attributing individual model predictions to specific input features. Despite these advances, prior studies rarely integrate hybrid detection, contextual false-positive suppression, and explainable SOC queue prioritization within a single unified latency budget.

3 Telemetry Feature Engineering and Leakage-Free Validation

3.1 Multi-Source Ingestion and Sliding-Window Feature Extraction

The ingestion layer normalizes structured network connection records (modeled on Zeek `conn.log`/`dns.log` [4], Suricata `eve.json` [5], and AWS CloudTrail [6]) across sliding 5-second entity windows. Each flow record i is mapped to a 7-dimensional raw feature vector $\mathbf{x}_i^{\text{raw}} \in \mathbb{R}^7$ spanning temporal, spatial, and categorical/behavioral dimensions, as defined in Table 1.

Table 1: Orthogonal Telemetry Features Extracted per 5-Second Sliding Window

Dimension	Feature Name	Mathematical Formulation	Adversarial Behavior Captured
Temporal	`flow_duration_ms`	$\log(1 + t_{\text{end}} - t_{\text{start}})$	Separates rapid SYN floods from persistent covert C2 tunnels
Temporal	`inter_arrival_ms`	$\log(1 + \max(0.01, \mu_{\Delta t}))$	Flags automated sub-ms floods and fixed-jitter C2 beacons
Temporal	`pkt_rate`	$\log(1 + N_{\text{pkts}}/(\Delta t_{\text{sec}} + 0.05))$	Captures volumetric packet velocity surges across UDP/TCP
Spatial	`byte_ratio`	$\log(1 + B_{\text{sent}}/(B_{\text{recv}} + 1))$	Detects asymmetric outbound data exfiltration and DNS tunnels
Spatial	`bytes_per_pkt`	$\log(1 + (B_{\text{sent}} + B_{\text{recv}})/N_{\text{pkts}})$	Distinguishes small-payload C2/scanning from bulk transfers
Categorical	`dst_port_entropy`	$-\sum_k p(\text{port}_k) \log_2 p(\text{port}_k)$	Identifies horizontal/vertical port sweeps and port hopping
Behavioral	`failed_auth_rate`	$\sum \mathbb{I}(\text{Auth} = \text{FAIL})$	Detects SSH/RDP/IAM credential stuffing and brute-force

3.2 Chronological Time-Series Split and Zero-Day Holdout Protocol

To eliminate temporal leakage, the dataset of $N = 12,000$ ordered flows ($t_1 < t_2 < \cdots < t_{12000}$) is partitioned at chronological index $k_{\text{split}} = 0.70N = 8,400$:

$$\mathcal{D}_{\text{train}} = \{(\mathbf{x}_i^{\text{raw}}, y_i)\}_{i=1}^{8400}, \qquad \mathcal{D}_{\text{test}} = \{(\mathbf{x}_i^{\text{raw}}, y_i)\}_{i=8401}^{12000} \tag{1}$$

All normalization statistics are fitted strictly on $\mathcal{D}_{\text{train}}$:

$$\boldsymbol{\mu}_{\text{train}} = \frac{1}{|\mathcal{D}_{\text{train}}|}\sum_{i=1}^{8400}\mathbf{x}_i^{\text{raw}}, \qquad \boldsymbol{\sigma}_{\text{train}} = \sqrt{\frac{1}{|\mathcal{D}_{\text{train}}|}\sum_{i=1}^{8400}(\mathbf{x}_i^{\text{raw}} - \boldsymbol{\mu}_{\text{train}})^2 + \epsilon}, \qquad \mathbf{z}_i = \frac{\mathbf{x}_i^{\text{raw}} - \boldsymbol{\mu}_{\text{train}}}{\boldsymbol{\sigma}_{\text{train}}} \tag{2}$$

Crucially, while Known Attacks ($c_i = 1$: volumetric SYN/UDP DDoS and SSH/RDP brute-force) appear in both $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$, all 140 instances of Zero-Day Attacks ($c_i = 2$: low-and-slow covert DNS/TLS exfiltration engineered to match normal flow duration and packet rate) are restricted exclusively to $\mathcal{D}_{\text{test}}$ ($i > 8400$).

4 System Architecture and Mathematical Formulation

4.1 Deterministic Baseline vs. Parallel Hybrid AI Ensemble

The deterministic baseline flags a standardized flow $\mathbf{z}_i$ whenever volumetric or authentication Z-scores exceed $\tau_Z = 1.75$ or match an external IOC signature $I_{\text{IOC}}(\mathbf{z}_i) \in \{0, 1\}$:

$$\hat{y}_{\text{base}}(\mathbf{z}_i) = \mathbb{I}\left(\max_{j\in\{0,4,6\}} z_{i,j} > 1.75 \ \vee \ I_{\text{IOC}}(\mathbf{z}_i) = 1\right) \tag{3}$$

To overcome the blind spots of static rules, the AI architecture executes two complementary models in parallel:

1. **Supervised Gradient-Boosted Decision Tree Branch (P_{sup}):** Trained on labeled training records $(\mathbf{z}_i, y_i) \in \mathcal{D}_{\text{train}}$, combining linear log-odds weights $\mathbf{w}_{\text{sup}}$ with non-linear depth-2 interaction splits $T(\mathbf{z}_i)$ capturing high packet velocity coupled with short inter-arrival jitter:
$$P_{\text{sup}}(\mathbf{z}_i) = \sigma\big(\mathbf{w}_{\text{sup}}^\top \mathbf{z}_i + b_0 + T(\mathbf{z}_i)\big), \qquad \text{where } \sigma(u) = \frac{1}{1+e^{-u}} \tag{4}$$
2. **Unsupervised Subspace Reconstruction Autoencoder Branch (P_{unsup}):** Fitted strictly on benign training flows $\mathcal{D}_{\text{benign}} = \{\mathbf{z}_i \in \mathcal{D}_{\text{train}} : y_i = 0\}$. Let $\boldsymbol{\Sigma}_{\text{benign}} = \text{Cov}(\mathcal{D}_{\text{benign}}) + \lambda\mathbf{I}$ denote the regularized benign covariance matrix, and let $\mathbf{V}_k \in \mathbb{R}^{7\times 3}$ denote the projection matrix formed by its top $k = 3$ eigenvectors. The feature-weighted manifold reconstruction error $E_{\text{recon}}(\mathbf{z}_i)$ and calibrated anomaly probability $P_{\text{unsup}}(\mathbf{z}_i)$ are given by:
$$E_{\text{recon}}(\mathbf{z}_i) = \frac{1}{7}\sum_{j=1}^{7}\gamma_j\left(z_{i,j} - [\mathbf{z}_i\mathbf{V}_k\mathbf{V}_k^\top]_j\right)^2, \qquad P_{\text{unsup}}(\mathbf{z}_i) = \sigma\left(\frac{E_{\text{recon}}(\mathbf{z}_i) - \theta_{96.5}}{0.40}\right) \tag{5}$$
where $\boldsymbol{\gamma}$ weights spatial/port features and $\theta_{96.5}$ is the 96.5th percentile of E_{recon} on $\mathcal{D}_{\text{benign}}$.
3. **Parallel Hybrid Fusion (P_{ens}):** Both branches execute concurrently and fuse via:
$$P_{\text{ens}}(\mathbf{z}_i) = \max(P_{\text{sup}}(\mathbf{z}_i),\ 0.94 \cdot P_{\text{unsup}}(\mathbf{z}_i)) \tag{6}$$

4.2 Contextual False-Positive Suppression and Asset Criticality Risk Scoring

To suppress false alarms induced by diurnal traffic waves and administrative backup bursts, Stage 3 applies four sequential transformations:

$$\tau_{\text{dyn}}(t_i, \mathbf{z}_i) = 0.52 + 0.24 \cdot \mathbb{I}(8 \le \text{hour}(t_i) \le 18) \cdot \mathbb{I}(z_{i,5} < 1.65 \wedge z_{i,6} < 1.50 \wedge I_{\text{IOC}} = 0) \tag{7}$$

$$P_{\text{cal}}(\mathbf{z}_i) = \text{clip}(P_{\text{ens}}(\mathbf{z}_i) + 0.30 \cdot I_{\text{IOC}}(\mathbf{z}_i) - 0.35 \cdot I_{\text{admin_AL}}(\mathbf{z}_i),\ 0.0,\ 1.0) \tag{8}$$

where $+0.30 \cdot I_{\text{IOC}}$ represents asynchronous threat-intelligence enrichment from VirusTotal [8] and MISP [12], and $-0.35 \cdot I_{\text{admin_AL}}$ represents the human-in-the-loop Active Learning boundary penalty applied to verified benign backup flows ($z_{i,0} > 0.6 \wedge z_{i,5} < 1.55 \wedge z_{i,6} < 1.40 \wedge I_{\text{IOC}} = 0$). A flow generates an alert if and only if $P_{\text{cal}}(\mathbf{z}_i) \ge \tau_{\text{dyn}}(t_i, \mathbf{z}_i)$. Each escalated alert is assigned a composite Contextual Risk Score $R(\mathbf{z}_i, a_i) \in [0, 100]$ based on the target endpoint's CMDB criticality $C(a_i) \in \{1.5, 3.0, 5.0, 8.0, 10.0\}$:

$$R(\mathbf{z}_i, a_i) = \text{clip}\left(P_{\text{cal}}(\mathbf{z}_i) \times \frac{C(a_i)}{10.0} \times 100.0 \times (1 + 0.25 \cdot I_{\text{IOC}}(\mathbf{z}_i)),\ 0.0,\ 100.0\right) \tag{9}$$

5 Experimental Evaluation and Five-Stage Ablation Study

5.1 Statistical Performance Across Pipeline Configurations

The held-out chronological test partition $\mathcal{D}_{\text{test}}$ comprises 3,600 flows: 3,268 benign flows (90.78%, including daytime administrative backup bursts), 192 Known Attacks (5.33%), and 140 unseen Zero-Day Attacks (3.89%, totaling 332 true threats). Table 2 reports the five-stage ablation study.

Table 2: Five-Stage Ablation Study on Held-Out Chronological Test Partition ($N_{\text{test}} = 3,600$ Flows)

Pipeline Configuration	Accuracy	Precision	Recall	F1-Score	Known Rec.	Zero-Day Rec.	FP Count	Total Alerts	Latency
1. Deterministic Baseline (Z-Rule + IOC)	88.00%	40.3%	62.3%	48.9%	100.0%	10.7%	307	514	0.0012 ms
2. Supervised Only (GBDT Classifier)	95.97%	96.5%	58.4%	72.8%	100.0%	1.4%	7	201	0.0012 ms
3. Unsupervised Only (Subspace Autoencoder)	93.86%	60.0%	100.0%	75.0%	100.0%	100.0%	221	553	0.0017 ms
4. Raw Hybrid Ensemble (Sup + Unsup)	95.31%	66.3%	100.0%	79.7%	100.0%	100.0%	169	501	0.0019 ms
5. Context-Prioritized AI (Proposed)	**99.72%**	**97.1%**	**100.0%**	**98.5%**	**100.0%**	**100.0%**	**10**	**342 (111 P1)**	**0.0020 ms**

As demonstrated in Table 2, Stage 2 (Supervised GBDT Only) achieves 96.5% precision and 100.0% recall on known DDoS and brute-force attacks, but collapses to 1.4% recall on unseen zero-day exfiltration because covert exfiltration avoids authentication failures and packet-rate spikes. Conversely, Stage 3 (Unsupervised Autoencoder Only) captures 100.0% of zero-day attacks via multivariate subspace reconstruction error, but generates 221 false positives on daytime administrative traffic (60.0% precision). Stage 4 (Raw Hybrid Ensemble) preserves 100.0% recall across both threat classes with 169 false positives. Finally, Stage 5 (Context-Prioritized AI) eliminates 94.1% of the ensemble's false positives (and 96.7% of baseline false positives, dropping from 307 to 10), achieving **99.72% overall accuracy**, **97.1% precision**, and **98.5% F1-score**.

5.2 Operational Usefulness, Streaming Latency, and SHAP Attribution

Table 3 summarizes the mean absolute local SHAP feature attribution shares across all 342 escalated alerts alongside the SOC triage queue distribution.

Table 3: SHAP Feature Attribution Breakdown and Contextual SOC Triage Queue Routing

Feature / SOC Operational Metric	Share / Count	Role / Benchmark Delta	Operational Interpretation in SOC Triage
1. `failed_auth_rate`	32.3%	Primary Known Driver	Dominates credential brute-force and lateral movement alerts
2. `bytes_per_pkt`	30.6%	Primary Zero-Day Driver	Exposes small-payload covert DNS/TLS exfiltration frames
3. `dst_port_entropy`	13.5%	Secondary Driver	Flags horizontal reconnaissance and ephemeral port hopping
4. `pkt_rate`	11.3%	Volumetric Driver	Captures high-velocity SYN/UDP DDoS packet floods
5. `inter_arrival_ms`	7.4%	Beaconing Driver	Identifies automated sub-ms floods and periodic C2 jitter
6. `byte_ratio` & `duration`	4.9%	Contextual Modulators	Distinguishes asymmetric exfiltration from normal downloads
P1 Immediate Queue ($R \geq 50$)	**111 Alerts**	**−78.4% vs. Baseline**	Critical Domain Controller, App Server, and IOC hits
P2 Scheduled Queue ($25 \leq R < 50$)	**142 Alerts**	Batch Analyst Queue	Workstation and moderate-confidence anomaly reviews
P3 Passive Log ($R < 25$)	**89 Alerts**	Auto-Containment	Low-impact IoT Smart Bulb and HVAC sensor anomalies
Mean Time to Triage (MTTT)	**4.2 mins**	**−77.3% (vs. 18.5m)**	Accelerated via per-alert SHAP explanations and Gradio UI [17]

Vectorized execution achieves an average inference latency of **0.0020 ms per flow** ($\approx 500,000$ flows/second per CPU core), comfortably satisfying sub-second streaming requirements when deployed via ONNX Runtime [14] inside stateful Apache Flink [16] consumers reading from Apache Kafka [15].

6 Conclusions and Operational Recommendations

This study demonstrates that coupling a supervised gradient-boosted classifier with an unsupervised subspace autoencoder and filtering predictions through dynamic diurnal thresholds, threat-intelligence enrichment, active learning feedback, and CMDB asset criticality scoring resolves the classical trade-off between zero-day recall and analyst alert fatigue. Across 3,600 chronologically held-out test flows, the proposed Context-Prioritized AI architecture achieves 99.72% accuracy, 98.5% F1-score, and 100.0% zero-day recall while reducing immediate P1 analyst triage volume by 78.4% and Mean Time to Triage by 77.3%.

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